

JEE MAIN REVISION MASTER: QUADRATIC EQUATION AND INEQUALITIES

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 196 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Roots of Quadratic & Vieta's Relations	181 PYQs	Quadratic Roots
Very High	Location of Roots & Quadratic Inequalities	15 PYQs	Location of Roots

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Roots of Quadratic & Vieta's Relations	$\alpha + \beta = -\frac{b}{a}$, $\alpha\beta = \frac{c}{a}$, $ \alpha - \beta = \frac{\sqrt{D}}{ a }$, $aS_n + bS_{n-1} + cS_{n-2} = 0$ (Newton)
Location of Roots & Quadratic Inequalities	$\text{Both roots } > k \iff D \geq 0, -\frac{b}{2a} > k, a \cdot f(k) > 0$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Roots of Quadratic & Vieta's Relations (Very High Yield — 181 PYQs)

Core Principle: Discriminant analysis for real/complex roots, symmetric root functions, and Newton's power sum theorem.

■ **Caution / JEE Pitfall:** If coefficients a, b, c are rational and D is not a perfect square, roots occur in irrational conjugate pairs $(p \pm \sqrt{q})$.

■ **Speed Trick:** Newton's theorem: If α, β are roots of $ax^2 + bx + c = 0$ and $S_n = \alpha^n + \beta^n$, then $aS_n + bS_{n-1} + cS_{n-2} = 0$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $\lim_{x \rightarrow 2} \frac{\tan(x-2)}{(x^2+(p-2)x-2p)(x-2)^2} = 5$ for some $r, p \in \mathbb{R}$. If the set of all possible values of p is $[r, \infty)$, then $r =$?" [Ans: C]
- [2026 Morning Shift] Q#1 (MCQ): "Let $\alpha, \alpha+2, \alpha \in \mathbb{Z}$, be the roots of the quadratic equation $x(x+2)+(x+1)(x+3)+(x+2)(x+4)+\dots+(x+n-1)(x+n+1)=4n$ for some $n \in \mathbb{N}$." [Ans: C]
- [2026 Morning Shift] Q#1 (MCQ): "The number of the real solutions of the equation: $x|x+3|+|x-1|-2=0$ is..." [Ans: N/A]

3.2 Location of Roots & Quadratic Inequalities (Very High Yield — 15 PYQs)

Core Principle: Conditions for roots relative to real numbers k , k_1 , k_2 , and signs of quadratic expression $ax^2 + bx + c$.

■ **Caution / JEE Pitfall:** For both roots to be greater than k , checking $a \cdot f(k) > 0$ alone is NOT sufficient; you must ALSO check $D \geq 0$ and $-(b/2a) > k$.

■ **Speed Trick:** If $a > 0$, the quadratic $f(x) = ax^2 + bx + c > 0$ for all real x if and only if $D < 0$.

Representative Source PYOs:

- **[2025 Evening Shift] Q#1 (Numerical):** "If the set of all $a \in \mathbb{R}$, for which the roots of the equation $(1-a)x^2 + 2(a-3)x + 9 = 0$ are positive is $(-\infty, -\alpha] \cup [\beta, \infty)$ " **[Ans: 7]**
- **[2024 Evening Shift] Q#1 (Numerical):** "The number of real solutions of the equation $x(x^2 + 3|x| + 5|x-1| + 6|x-2|) = 0$ is _____." **[Ans: 1]**
- **[2023 Evening Shift] Q#3 (Numerical):** "Let m and n be the numbers of real roots of the quadratic equations $x^2 - 12x + |x| + 31 = 0$ and $x^2 - 5|x + 2| - 4 = 0$ respectively, where $|x| \dots$ " **[Ans: 9]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Evening Shift (Q#1)

Let $\lim_{x \rightarrow 2} \frac{(\tan(x-2))(rx^2+(p-2)x-2p)(x-2)^2}{(x-2)^2} = 5$ for some $r, p \in \mathbb{R}$. If the set of all possible values of q , such that the roots of the equation $rx^2-px+q=0$ lie in $(0, 2)$, be the interval $(\alpha, \beta]$, then $4(\alpha+\beta)$ equals :

- (A) 11
- (B) 13
- (C) 17 **[Correct]**
- (D) 21

Correct Answer: C

Solution Analysis: We are given $\lim_{x \rightarrow 2} \frac{\tan(x-2)(rx^2+(p-2)x-2p)}{(x-2)^2} = 5$. We have to first find the relation between r and p , and then determine all values of q for which the roots of $rx^2-px+q=0$ lie in $(0,2)$. Step 1: Simplify the limit Let $f(x)=rx^2+(p-2)x-2p$. Then the given limit becomes $\lim_{x \rightarrow 2} \frac{\tan(x-2)f(x)}{(x-2)^2} = 5$. Now, as $x \rightarrow 2$, $\tan(x-2) \sim x-2$. So the limit becomes $\lim_{x \rightarrow 2} \frac{f(x)}{x-2} = \lim_{x \rightarrow 2} \frac{f(x)}{x-2}$. For this limit to be finite,

Benchmark Problem 2: JEE Main 2025 (Online) 2nd April Evening Shift (Q#1)

If the set of all $a \in \mathbb{R}$, for which the roots of the equation $(1-a)x^2 + 2(a-3)x + 9 = 0$ are positive is $(-\infty, -\alpha] \cup [\beta, \gamma)$, then $2\alpha + \beta + \gamma$ is equal to .

Correct Answer: 7

Solution Analysis: $f(x)=(1-a)x^2+2(a-3)x+9$, $f(0)=9>0 \Rightarrow D \geq 0 \Rightarrow 4(a-3)^2 \geq 4(1-a) \cdot 9 \Rightarrow a \in (-\infty,-3] \cup [0,\infty)$ (i)
 $x_1+x_2=(-2(a-3)/1-a)$, $x_1x_2=(9/1-a)$ $x_1+x_2 > 0 \Rightarrow (a-3)/(1-a) > 0 \Rightarrow a \in (-\infty,1) \cup (3,\infty)$... (ii) $x_1x_2 > 0 \Rightarrow 1-a > 0 \Rightarrow a \in (-\infty,1)$... (iii) $\begin{aligned} &\& \text{\textit{Interaction of (i), (ii) and (iii)}} \\ &\& \end{aligned}$ $a \in (-\infty,-3] \cup [0,1)$